

The physical origin of $i\lambda c_i^\dagger c_j$ is various:

- (i) Peierls phase (gauge transformation)
- (ii) Time-reversal symmetry breaking due to magnetism
- (iii) Self-consistent order from the mean-field treatment of intersite Coulomb interaction

In this note, one focuses on the third scenario. First, one starts with the extended Hubbard interaction:

$$H_V = \sum_{\langle i,j \rangle} \sum_{\sigma\sigma'} V_{ij} n_{i\sigma} n_{j\sigma'}$$

where i and j are lattice sites, σ and σ' denote spin (or orbital) index, V_{ij} is the interaction matrix. Writing more explicit form, the Hamiltonian is expressed as:

$$H_V = \sum_{\langle i,j \rangle} \sum_{\sigma\sigma'} V_{ij} c_{i\sigma}^\dagger c_{i\sigma} c_{j\sigma'}^\dagger c_{j\sigma'}$$

Neglecting superconducting anomalous contraction, such as $\langle c_i c_j \rangle$ and $\langle c_i^\dagger c_j^\dagger \rangle$, the mean-field approximation results in:

$$\begin{aligned} \langle c_{i\sigma}^\dagger c_{i\sigma} c_{j\sigma'}^\dagger c_{j\sigma'} \rangle &\approx \langle c_{i\sigma}^\dagger c_{i\sigma} \rangle c_{j\sigma'}^\dagger c_{j\sigma'} + c_{i\sigma}^\dagger c_{i\sigma} \langle c_{j\sigma'}^\dagger c_{j\sigma'} \rangle - \langle c_{i\sigma}^\dagger c_{i\sigma} \rangle \langle c_{j\sigma'}^\dagger c_{j\sigma'} \rangle \\ &\quad - \langle c_{i\sigma}^\dagger c_{j\sigma'} \rangle c_{j\sigma'}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger c_{j\sigma'} \langle c_{j\sigma'}^\dagger c_{i\sigma} \rangle + \langle c_{i\sigma}^\dagger c_{j\sigma'} \rangle \langle c_{j\sigma'}^\dagger c_{i\sigma} \rangle \end{aligned}$$

One can define $\langle c_{i\sigma}^\dagger c_{i\sigma} \rangle$ and $\langle c_{i\sigma}^\dagger c_{j\sigma'} \rangle$ as density order parameter $\rho_{i\sigma}$ and bond order parameter $\chi_{ij}^{\sigma\sigma'}$, respectively. Rewriting the equation above,

$$\begin{aligned} \langle c_{i\sigma}^\dagger c_{i\sigma} c_{j\sigma'}^\dagger c_{j\sigma'} \rangle &\approx \rho_{i\sigma} c_{j\sigma'}^\dagger c_{j\sigma'} + c_{i\sigma}^\dagger c_{i\sigma} \rho_{j\sigma'} - \rho_{i\sigma} \rho_{j\sigma'} \\ &\quad - \chi_{ij}^{\sigma\sigma'} c_{j\sigma'}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger c_{j\sigma'} \chi_{ji}^{\sigma'\sigma} + \chi_{ij}^{\sigma\sigma'} \chi_{ji}^{\sigma'\sigma} \end{aligned}$$

The summation over σ and σ' yields:

$$\begin{aligned} \langle n_i n_j \rangle &\approx \langle n_i \rangle n_j + n_i \langle n_j \rangle - \langle n_i \rangle \langle n_j \rangle \\ &\quad - \sum_{\sigma\sigma'} \left[\chi_{ij}^{\sigma\sigma'} c_{j\sigma'}^\dagger c_{i\sigma} + c_{i\sigma}^\dagger c_{j\sigma'} \chi_{ji}^{\sigma'\sigma} - \chi_{ij}^{\sigma\sigma'} \chi_{ji}^{\sigma'\sigma} \right] \end{aligned}$$

where $n_i = \sum_{\sigma} c_{i\sigma}^\dagger c_{i\sigma}$. The first three terms are nothing but Hartree term. The rest of it (a.k.a Fock term) can be written as the Hamiltonian form:

$$H_V^{MF} = - \sum_{\langle i,j \rangle} V_{ij} \sum_{\sigma\sigma'} \left[\chi_{ij}^{\sigma\sigma'} c_{j\sigma'}^\dagger c_{i\sigma} + \text{h.c.} \right] + \sum_{\langle i,j \rangle} V_{ij} \sum_{\sigma\sigma'} \chi_{ij}^{\sigma\sigma'} \chi_{ji}^{\sigma'\sigma}$$

The last term with $\chi_{ij}^{\sigma\sigma'} \chi_{ji}^{\sigma'\sigma}$ corrects the double counting of the mean-field

energy. The first term can be coupled to the single-particle Hamiltonian H_0 . One will call it the effective interacting Hamiltonian H^{eff} :

$$H_0 = \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} h_{ij}^{\sigma\sigma'} c_{i\sigma}^\dagger c_{j\sigma'} + \text{h.c.}$$

$$H^{\text{eff}} = \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (h_{ij}^{\sigma\sigma'} - V_{ij} \chi_{ij}^{\sigma\sigma'}) c_{i\sigma}^\dagger c_{j\sigma'} + \text{h.c.}$$

Assuming spin-independent case for simplicity, the indices σ and σ' disappear. Furthermore, one should note that χ_{ij} is generally not "real." (This is because an arbitrary gauge choice $c_i \rightarrow e^{i\theta_i} c_i$ does not cancel out $c_j^\dagger \rightarrow e^{-i\theta_j} c_j^\dagger$.) Then χ_{ij} can be expressed as:

$$\chi_{ij} = \text{Re}[\chi_{ij}] + i \text{Im}[\chi_{ij}]$$

Accordingly, H^{eff} is given by:

$$H_{\text{spinless}}^{\text{eff}} = \sum_{\langle ij \rangle} (h_{ij} - V_{ij} \text{Re}[\chi_{ij}]) c_i^\dagger c_j - i \sum_{\langle ij \rangle} V_{ij} \text{Im}[\chi_{ij}] c_i^\dagger c_j$$

Thus, the conclusion one can make here is that "the mean-field treatment of intersite Coulomb interaction can result in the complex hopping". More precisely, the complex hopping term can be derived by the Fock channel of intersite Coulomb interaction with nonzero bond order parameter χ_{ij} .